Microeconomics I

Class 6 — Expenditure minimisation and the cost of living

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Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Short quiz
10:10	Warm-up; two price indices
10:20	Expenditure minimisation: h and e
10:31	In pairs: kinks and corners
10:38	Relabelling; properties of e and h
10:50	Break
11:00	Shephard's lemma; duality
11:11	Compensated price effects; net substitutes
11:19	The true cost of living
11:25	In pairs: fixed basket vs true index; summary

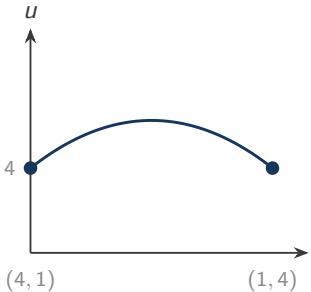
Practicalities

- Problem set 1: on the desk at the start of class, one paper copy per group
- Problem set 2: handed out in class today, one per group;
  due at the start of class, Wed 21 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

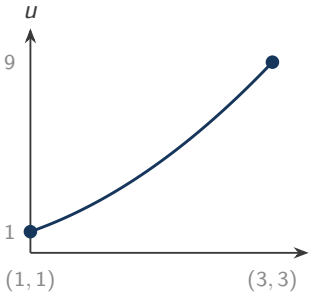
  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: concave, or only quasiconcave?

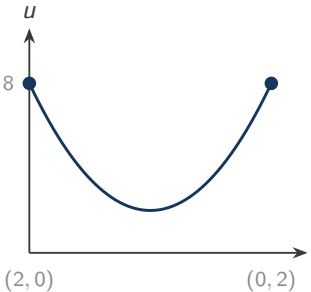
Utility along the segment between two bundles. What does each picture rule out?



x_1x_2 **from** $(4,1)$ **to** $(1,4)$



x_1x_2 **from** $(1,1)$ **to** $(3,3)$

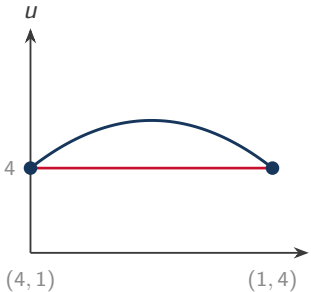


$x_1^3 + x_2^3$ **from** $(2,0)$ **to** $(0,2)$

Class 3, slides 21–22; MWG 3.C, pp. 49–50.

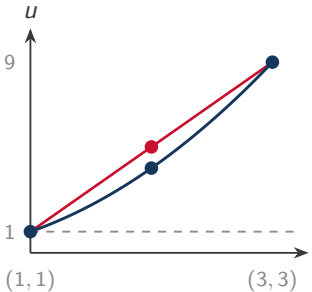
Warm-up: concave, or only quasiconcave?

Utility along the segment between two bundles. What does each picture rule out?



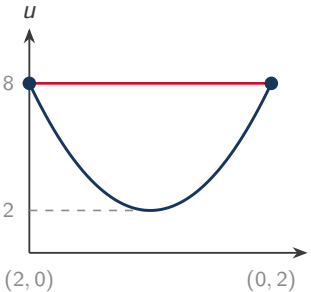
x_1x_2 **from** $(4,1)$ **to** $(1,4)$

nothing ruled out



x_1x_2 **from** $(1,1)$ **to** $(3,3)$

not concave



$x_1^3 + x_2^3$ **from** $(2,0)$ **to** $(0,2)$

not quasiconcave

Concave: never below the chord. Quasiconcave: never below the lower end.

Class 3, slides 21–22; MWG 3.C, pp. 49–50.

Two US price indices, the same prices: which rises faster?

United States: two official price indices, the same prices

CPI-U: today's prices of an earlier basket (Laspeyres)

Chained CPI-U, since 2002: the basket follows what people buy

Which rises faster, and by how much a year?

0.02 · 0.2 · 2 percentage points (average, 2001–2023)

U.S. Bureau of Labor Statistics: CPI-U and Chained CPI-U documentation.

The earlier basket runs ahead: about 0.2 points a year

CPI-U, earlier basket

+0.2

points a year above the chained CPI-U (average, Dec. to Dec., 2001–2023)

since 2017

tax brackets

US federal brackets rise with the chained CPI-U

Why a fixed basket runs ahead of the true cost of living: after the break.

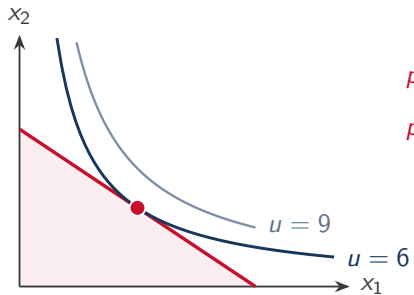
BLS, Chained CPI: questions and answers (bls.gov/cpi); BLS, Introducing the Chained CPI.

Expenditure minimisation

MWG 3.E

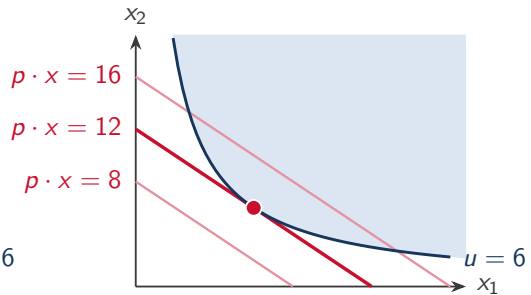
Two problems, one tangency: fix the budget, or fix the curve

EMP: $\min_{x \geq 0} \; p \cdot x \;$ s.t. $\; u(x) \geq u$ value $e(p, u)$, solution $h(p, u)$



UMP: $w = 12$ **fixed**

the highest curve the line reaches: $u = 6$



EMP: $u = 6$ **fixed**

the lowest line that reaches the curve: 12

Same bundle $(3, 2)$: the best bundle 12 euros buy is the cheapest way to reach $u = 6$.

MWG 3.E, p. 58: Figure 3.E.1, Proposition 3.E.1; $u = x_1 x_2$, $p = (2, 3)$.

For x_1x_2: compensated demand and the least cost of u

tangency $x_2/x_1 = p_1/p_2$, on the curve $x_1x_2 = u$:

$$h(p,u) = \left(\sqrt{u\,p_2/p_1},\ \sqrt{u\,p_1/p_2}\right)$$

$$e(p,u) = p \cdot h(p,u) = 2\sqrt{u\,p_1p_2}$$

at $p = (2,3)$, $u = 6$: $h = (3,2)$, $e = 12$

No w anywhere: h and e depend on prices and the target u.

MWG 3.E, pp. 59–61; Example 3.E.1, p. 63, does $x_1^{\alpha}x_2^{1-\alpha}$; ours is x_1x_2.

e and v undo each other: four identities

identity	check for $x_1 x_2$
$e(p, v(p, w)) = w$	$2\sqrt{\dfrac{w^2}{4p_1p_2}\, p_1p_2} = w$
$v(p, e(p, u)) = u$	$\dfrac{e(p, u)^2}{4p_1p_2} = u$
$h(p, u) = x(p, e(p, u))$	$\dfrac{e(p, u)}{2p_1} = \sqrt{u\, p_2/p_1}$
$x(p, w) = h(p, v(p, w))$	$\sqrt{v(p, w)\, p_2/p_1} = \dfrac{w}{2p_1}$

Know v? Invert it in its last argument: that is e.

MWG Prop. 3.E.1, p. 58; (3.E.1), p. 60; (3.E.4), p. 62. $v = w^2/(4p_1p_2)$: class 5, slide 8.

In pairs: the cheapest way to the curve, with kinks and corners

$p = (2, 3)$: the cheapest bundle that reaches u; then $e(p, u)$ for any p.

$u(x)$	reach	cheapest bundle	$e(p, u)$
$\min\{x_1, x_2\}$	$u = 2.4$		
$x_1 + x_2$	$u = 6$		
$x_1^3 + x_2^3$	$u = 216$		

Four minutes.

In pairs: the cheapest way to the curve, with kinks and corners

$p = (2, 3)$: the cheapest bundle that reaches u; then $e(p, u)$ for any p.

$u(x)$	reach	cheapest bundle	$e(p, u)$	
$\min\{x_1, x_2\}$	$u = 2.4$	$(2.4,\ 2.4)$	$u\,(p_1 + p_2)$	the kink: she buys pairs
$x_1 + x_2$	$u = 6$	$(6,\ 0)$	$u \min\{p_1, p_2\}$	all on the cheaper good
$x_1^3 + x_2^3$	$u = 216$	$(6,\ 0)$	$u^{1/3} \min\{p_1, p_2\}$	a corner; the tangency is the dearest point

Four minutes.

Every row costs 12: $e(p, v(p, 12)) = 12$ again.

Relabel the curves: the cheapest bundle and its cost stay put

u	$v(p,w)$	$e(p,u)$	the curve through $(3,2)$
x_1x_2	$\dfrac{w^2}{4p_1p_2}$	$2\sqrt{u\,p_1p_2}$	$u=6$: $\;e=12$
$\frac{1}{2}\ln x_1+\frac{1}{2}\ln x_2$	$\ln w-\frac{1}{2}\ln(p_1p_2)-\ln 2$	$2\exp(u)\sqrt{p_1p_2}$	$u=\frac{1}{2}\ln 6$: $\;e=12$

Replace u by $f(u)$: v becomes $f(v)$; e and h keep euros and bundles, only the label changes.

MWG p. 57 ($\tilde{v}=f(v)$) and (3.E.1), p. 60; class 5, slide 12.

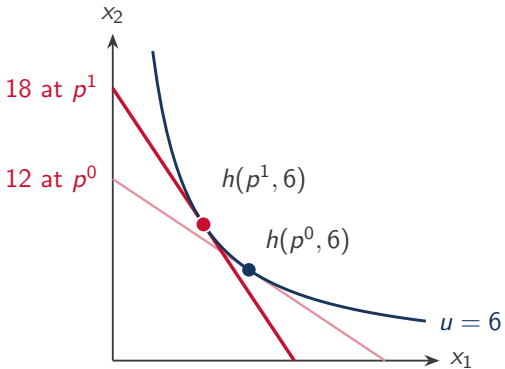
$e(p, u)$: homogeneous of degree 1 in p, increasing in u, concave in p

property	why	for $e = 2\sqrt{u\, p_1 p_2}$
(i) homogeneous of degree 1 in p	same curve, every cost $\times\, \alpha$	$2\sqrt{u\, \alpha p_1\, \alpha p_2} = \alpha\, e$
(ii) strictly increasing in u	continuity of u	$\dfrac{\partial e}{\partial u} = \sqrt{\dfrac{p_1 p_2}{u}} > 0$
nondecreasing in each p_ℓ	no bundle gets cheaper	$\dfrac{\partial e}{\partial p_1} = \sqrt{\dfrac{u\, p_2}{p_1}} > 0$
(iii) concave in p	after the break	
(iv) continuous		

Tuesday's v: degree 0 and quasiconvex. Today's e: degree 1 and concave.

MWG Proposition 3.E.2, p. 59: u continuous, $\succsim$ locally non-satiated, $X = \mathbb{R}^2_+$.

$h(p, u)$ stays on the curve and never rises with its own price



(i) $h(\alpha p, u) = h(p, u)$

only relative prices matter

(ii) $u\Big(h(p, u)\Big) = u$

no utility to spare

(iii) $\succsim$ strictly convex: one bundle

convex: a convex set of bundles

$(p^1 - p^0) \cdot \left(h^1 - h^0\right) \leq 0$

here $(2.5,\ 0) \cdot (-1,\ 1) = -2.5$

Good 1 dearer, utility held at $u = 6$: she buys less of it here, and never more in general.

MWG Prop. 3.E.3, p. 61; Prop. 3.E.4 and (3.E.5), p. 62; $u = x_1 x_2$, $p^0 = (2, 3)$, $p^1 = (4.5, 3)$.

(optional) Both prices move: which bundles can be the new h?

Utility target u fixed. At $p^0 = (2, 3)$ her cheapest bundle is $h^0 = (3, 2)$.
Prices move to $p^1 = (3, 1)$. Could $h(p^1, u)$ be this bundle?
Check slide 14's law and the definition of h, at both price vectors.

(1) $(2,\ 4)$

(2) $(4,\ 1)$

(3) $(2,\ 2.5)$

MWG Proposition 3.E.4 and (3.E.5), p. 62: u continuous, $\succsim$ locally non-satiated, $h(p, u)$ single-valued.

(optional) Both prices move: which bundles can be the new h?

Utility target u fixed. At $p^0 = (2, 3)$ her cheapest bundle is $h^0 = (3, 2)$.
Prices move to $p^1 = (3, 1)$. Could $h(p^1, u)$ be this bundle?
Check slide 14's law and the definition of h, at both price vectors.

(1) $(2,\ 4)$ yes: law $-5 \leq 0$; costs $16 \geq 12$ at p^0, $10 \leq p^1 \cdot h^0 = 11$ at p^1

(2) $(4,\ 1)$ no: law $(1, -2) \cdot (1, -1) = 3 > 0$

(3) $(2,\ 2.5)$ no, although the law holds (-2): at p^0 it costs $11.5 < e(p^0, u) = 12$

MWG Proposition 3.E.4 and (3.E.5), p. 62: u continuous, $\succsim$ locally non-satiated, $h(p, u)$ single-valued.

Break

Back at 11:00

Shephard's lemma and duality

MWG 3.F, 3.G

Shephard's lemma: compensated demand is the price slope of e

$$h_\ell(p,u) = \frac{\partial e(p,u)}{\partial p_\ell}$$

u continuous; $\succsim$ locally non-satiated and strictly convex

$e = 2\sqrt{u\, p_1 p_2}$: $\quad \dfrac{\partial e}{\partial p_1} = \sqrt{u\, p_2/p_1} = h_1$

slide 9

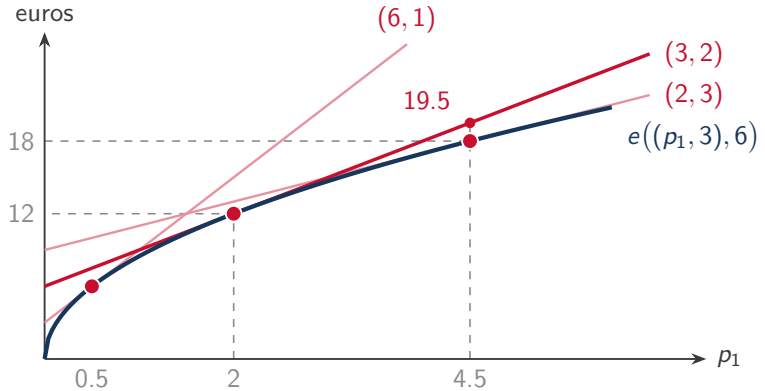
Roy (class 5): $\quad x_\ell = -\dfrac{\partial v/\partial p_\ell}{\partial v/\partial w}$ $\qquad\qquad$ Shephard: $\quad h_\ell = \dfrac{\partial e}{\partial p_\ell}$

Roy divides to turn utils into euros; e is in euros already.

MWG Proposition 3.G.1 (Shephard's lemma), p. 68; Roy's identity: Prop. 3.G.4, p. 73.

e is the lowest of straight lines: concave in p, slope $= h$

$u = x_1 x_2$, $u = 6$, $p_2 = 3$: each straight line is the cost $p_1 x_1 + 3x_2$ of one bundle on the curve



At $p_1 = 2$ the line of $(3, 2)$ touches e with slope $3 = h_1$: Shephard's lemma, drawn.

MWG Figure 3.E.2, p. 60; Figure 3.F.3 and the duality theorem (Prop. 3.F.1), pp. 65–66.

Compensated price effects: own effects ≤ 0, cross effects equal

$$D_p h(p,u) = \begin{pmatrix} \partial h_1/\partial p_1 & \partial h_1/\partial p_2 \\ \partial h_2/\partial p_1 & \partial h_2/\partial p_2 \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{3} \end{pmatrix}$$

$x_1 x_2$ at $p = (2,3)$, $u = 6$

own effects ≤ 0

negative semidefinite

$\partial h_1/\partial p_2 = \partial h_2/\partial p_1$

symmetric

$D_p h\ p = 0$: $\quad -0.75 \cdot 2 + 0.5 \cdot 3 = 0$

degree 0 in p

Two goods: $\partial h_1/\partial p_2 = -(p_1/p_2)\,\partial h_1/\partial p_1 \geq 0$.

MWG Proposition 3.G.2, p. 69: $D_p h(p,u) = D_p^2 e(p,u)$.

Net substitutes: with two goods, never net complements

net substitutes: $\partial h_1/\partial p_2 > 0$; net complements: < 0

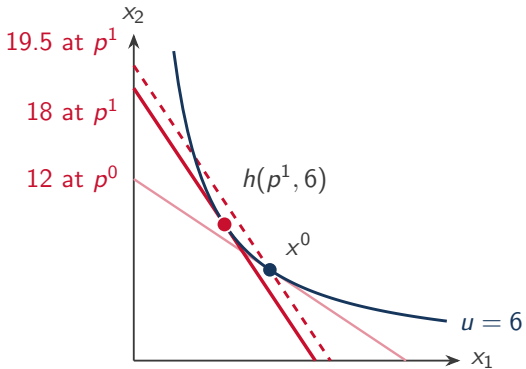
CES: $u = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho\right)^{1/\rho}$, $\sigma = 1/(1-\rho)$; $\sigma = 2$: same $\succsim$ as $\sqrt{x_1} + \sqrt{x_2}$

σ	u	gross: x_1, p_2: $3 \to 4$	net: h_1, p_2: $3 \to 4$
	$p_1 = 2$; gross at $w = 12$ (class 5); net at the utility $w = 12$ bought		
0	$\min\{x_1, x_2\}$	$2.4 \to 2$: complements	$2.4 \to 2.4$: neither
1	$x_1 x_2$	$3 \to 3$: neither	$3 \to 3.46$: substitutes
2	$\sqrt{x_1} + \sqrt{x_2}$	$3.6 \to 4$: substitutes	$3.6 \to 4.44$: substitutes

Gross and net can disagree; the gap is a wealth effect: next Tuesday.

MWG 3.G, p. 70 (weak inequalities; strict here, so zero is "neither"). Class 5, slide 23; CES: class 3, slide 26.

True cost of living: $\times 1.5$; the fixed basket says $\times 1.625$



true index: $\dfrac{e(p^1, u^0)}{e(p^0, u^0)} = \dfrac{18}{12} = 1.5$

fixed basket (Laspeyres):

$$\frac{p^1 \cdot x^0}{p^0 \cdot x^0} = \frac{19.5}{12} = 1.625$$

x^0 still reaches u^0, so $e(p^1, u^0) \leq p^1 \cdot x^0$: never below the truth

To stay on her curve she needs 6 more euros, not 7.5.

Konüs (1924; Econometrica 1939): the true index. Own numbers: $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.

In pairs: good 1 now costs 8: how much richer must each be?

$p^0 = (2, 3)$, $w = 12 \;\rightarrow\; p^1 = (8, 3)$. Fixed-basket index and true index.

$u(x)$	x^0	fixed basket $\frac{p^1 \cdot x^0}{12}$	true $\frac{e(p^1, u^0)}{12}$
$\min\{x_1, x_2\}$	(2.4, 2.4)		
$x_1 x_2$	(3, 2)		
$x_1 + x_2$	(6, 0)		

Seven minutes.

In pairs: good 1 now costs 8: how much richer must each be?

$p^0 = (2, 3)$, $w = 12 \;\rightarrow\; p^1 = (8, 3)$. Fixed-basket index and true index.

$u(x)$	x^0	fixed basket $\frac{p^1 \cdot x^0}{12}$	true $\frac{e(p^1, u^0)}{12}$	
$\min\{x_1, x_2\}$	$(2.4,\ 2.4)$	2.2	2.2	no substitution, no bias
$x_1 x_2$	$(3,\ 2)$	2.5	2	$h(p^1, 6) = (1.5,\ 4)$
$x_1 + x_2$	$(6,\ 0)$	4	1.5	switch to good 2: $(0,\ 6)$

Seven minutes.

The more she can substitute, the more the fixed basket overstates.

(optional) Price her new basket: how does that index compare?

Slide 22's consumer: $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.
At p^1 she buys x^1, with utility u^1.

(1) x^1, and the Paasche index $P = \dfrac{p^1 \cdot x^1}{p^0 \cdot x^1}$.

(2) Show, under the assumptions of
Prop. 3.E.1: $P \leq \dfrac{e(p^1, u^1)}{e(p^0, u^1)}$.

(3) Order P and slide 22's two indices.

(optional) Price her new basket: how does that index compare?

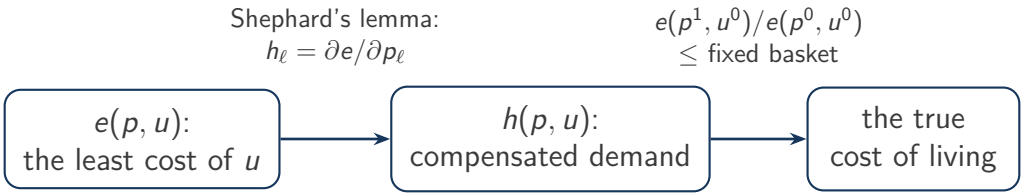
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At p^1 she buys x^1, with utility u^1.

(1) x^1, and the Paasche index $P = \dfrac{p^1 \cdot x^1}{p^0 \cdot x^1}$. $x^1 = (4/3,\ 2)$; $P = 18/13 \approx 1.38$

(2) Show, under the assumptions of Prop. 3.E.1: $P \leq \dfrac{e(p^1, u^1)}{e(p^0, u^1)}$. $e(p^1, u^1) = 12 = p^1 \cdot x^1$ (slide 10); x^1 reaches u^1, so $e(p^0, u^1) \leq p^0 \cdot x^1$

(3) Order P and slide 22's two indices. $P \approx 1.38 < 1.5$ (true) < 1.625 (fixed basket)

Today in one line



Tuesday: the Slutsky equation and the Slutsky matrix; summary of the relationships; compensating and equivalent variation.

Problem set 2: handed out in class today, one per group; due at the start of class, Wednesday 21 October, on paper.

Read MWG 3.G and 3.I (or Muñoz-Garcia ch. 2) for Tuesday 20 October.